When Is a Task Library Enough? Coverage and In-Context Forecasting

Ivan Habib
ivan.habib4@gmail.com

Abstract

When a sequence model forecasts a system it has never seen, it may be estimating that system's dynamics from the context or retrieving a similar system from its training set. The difference matters for time-series foundation models applied zero-shot to new series. We make the question testable one context at a time. Call Bayesian prediction restricted to the training tasks *lookup*. A classical importance-sampling quantity gives the number of training tasks lookup would need to predict well on a context; the actual library size divided by this number is the library's *coverage* of that context. A preregistered AR(1) experiment confirms that coverage predicts the error of lookup, including against a shuffled control with the same total number of tasks. A Transformer trained on 4,096 coupled eight-dimensional linear systems then forecasts the longest contexts within 0.1% of Bayes-optimal risk, although lookup would need about 115 times as many systems, and its predictions are 73 times closer (in squared distance) to the Bayes-optimal forecast than to lookup. On in-context linear regression, a Transformer comes within 5% of Bayes risk while a GRU of similar size stays 30% above it, although both are much closer to Bayes-optimal than to lookup.

1 Did the model infer or retrieve?

A temporal forecaster can be right for two quite different reasons: it may recognize systems represented during training and interpolate among them, or it may use the observed history to estimate dynamics that were never stored. When the training library is dense, both accounts can produce nearly the same forecast, so predictive accuracy alone does not tell us which kind of adaptation occurred.

The number of training tasks is a poor proxy for whether the library contains systems close to the one that generated a particular history: a long history can rule out every training system, while a short one may leave many plausible. This matters because time-series foundation models are trained on broad mixtures of real and synthetic series and then applied zero-shot to new ones [Das et al., 2024, Ansari et al., 2024]. Retrieval is a tempting explanation for their adaptability, but a large training set does not show that systems like the test system were in it.

Controlled studies have shown that Transformers can learn to fit unseen functions in context [Garg et al., 2022], that their in-context predictions can match gradient descent or ridge regression [Akyürek et al., 2023, von Oswald et al., 2023], and that in-context learning can be read as implicit Bayesian inference [Xie et al., 2022]. The training distribution still matters. In in-context regression, Raventós et al. [2023] show that as the number of training tasks grows, the model's predictions move from Bayes restricted to the training tasks toward ridge regression, and Nguyen and Reddy [2025] explain this transition by the different rates at which the two solutions are learned. Both describe the model's behavior as a function of library size. We ask a per-context question instead: for this particular history, was the training library large enough that lookup could explain the forecast? Classical importance-sampling identities [Chatterjee and Diaconis, 2018, Agapiou et al., 2017] answer it

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Foundation Models for Temporal Systems (FMTS).

exactly when the prior and likelihood are known. The identities are standard; what we add is using them as a test on a trained model's actual library, first in a controlled AR(1) experiment and then for Transformers trained on fixed libraries. A GRU trained on the same data tests whether moving away from lookup is enough to learn the Bayes-optimal rule.

2 Lookup, Bayes, and how large a library must be

Setup. Each task is a dynamical system with parameters θ drawn from a prior P. A training library is a set of M systems $\theta_1, \ldots, \theta_M$ drawn independently from P; the model is trained only on sequences generated by these M systems. At test time the model sees a context C, a history generated by a new system drawn from P. We write $L_C(\theta)$ for the likelihood of C under system θ, and $Q_C(\theta) \propto L_C(\theta)P(\theta)$ for the posterior.

Two reference predictors. *Continuous Bayes* is the Bayes-optimal forecast under P: it averages each system's forecast over the posterior Q_C. *Library Bayes* does the same computation over the M training systems only, giving system j the weight $w_j \propto L_C(\theta_j)$; in regression this is the discrete minimum mean-squared-error (dMMSE) estimator of Raventós et al. [2023]. A model that recognizes its training systems and combines them by posterior weight would behave like library Bayes.

How large must the library be? Library Bayes is self-normalized importance sampling: samples from one distribution (here the prior) are reweighted to approximate another (the posterior). How many samples it needs depends on

$$N_2(C) = \frac{\int L_C(\theta)^2 \, dP(\theta)}{\left(\int L_C(\theta) \, dP(\theta)\right)^2} = e^{D_2(Q_C \| P)} = 1 + \chi^2(Q_C \| P), \tag{1}$$

where the denominator is the squared marginal likelihood of C, written $(\mathbb{E}_P L_C)^2$ below, D_2 is the Rényi divergence of order two, and χ^2 is the chi-squared divergence. Two standard identities show what N_2 measures (Appendix A.1). Conditional on C,

$$\frac{\mathrm{Var}\left[M^{-1}\sum_j L_C(\theta_j)\right]}{\left(\mathbb{E}_P L_C\right)^2} = \frac{N_2(C) - 1}{M}, \qquad \frac{\mathrm{ESS}_M}{M} \xrightarrow{a.s.} \frac{1}{N_2(C)}, \tag{2}$$

where $\mathrm{ESS}_M = (\sum_j w_j)^2 / \sum_j w_j^2$ is the usual effective sample size of the library weights and the arrow denotes almost-sure convergence as M grows. So for large M, only about $M/N_2(C)$ training systems carry real weight for this context. We call $R(C) = M/N_2(C)$ the library's *coverage* of C; it is the large-library limit of the effective sample size, and we use N_2 because it can be computed exactly for each context without drawing a library. We also report the shortfall in bits, $G(C) = \log_2 N_2(C) - \log_2 M$, the number of times the library would have to double to reach $R = 1$. The error of library Bayes in estimating the posterior mean of any bounded function scales as $O(N_2/M)$ [Agapiou et al., 2017], so when G is several bits, lookup cannot be accurate.

A trained network is not bound by this limit, because it can share computation across tasks. So if a model matches continuous Bayes at a context where G is large, it cannot be doing lookup over its training systems. To measure which reference a model resembles, let Δ_D and Δ_C be the mean squared distances between the model's predictions and those of library Bayes and continuous Bayes. A distance ratio Δ_D/Δ_C above one, with a bootstrap interval excluding one, means the model is closer to continuous Bayes; a ratio near one means the two explanations fit equally well. Risk always means next-step mean squared error.

3 Coverage predicts the error of lookup

Setup. This experiment involves no trained model; it checks that coverage predicts the error of library Bayes. We use the stable AR(1) family $x_{t+1} = ax_t + \epsilon_t$, with $a \sim \mathrm{Unif}[-0.95, 0.95]$, $\epsilon_t \sim \mathcal{N}(0,1)$, and $x_0 = 0$. For each context length $n \in \{16, 64, 256, 1024\}$ observed transitions, we simulate 256 histories and compute $N_2(C)$ exactly by quadrature. For every history and each target coverage $R \in \{1/4, 1/2, 1, 2, 4, 8, 16, 32\}$, we set $M(C) = \lceil RN_2(C) \rceil$, so histories with narrower posteriors receive larger libraries, and average over 32 independently drawn libraries. The

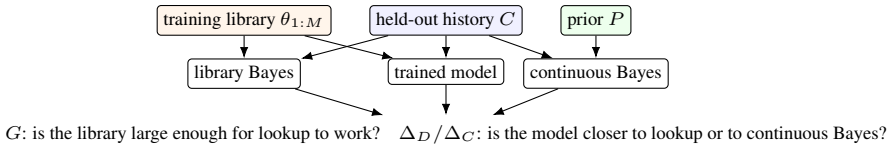


Figure 1: **The coverage test.** All three predictors see the same context; library Bayes and the model use the same training systems.

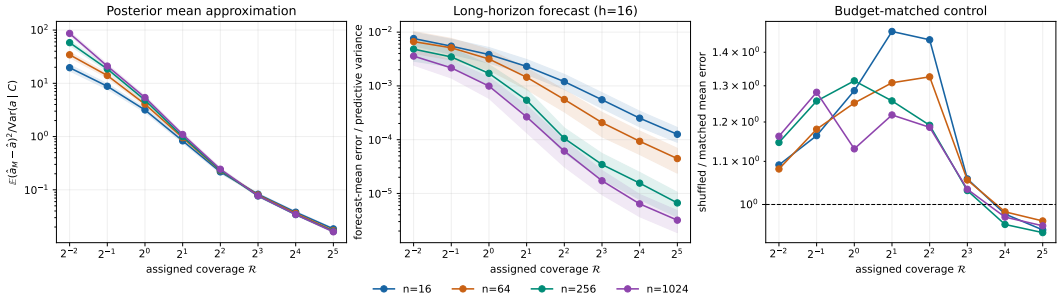


Figure 2: **Assigning libraries by coverage controls the error of lookup.** Each library has $M = \lceil RN_2(C) \rceil$ systems. Left: squared error of the library's posterior mean $\hat{a}_M$ relative to the exact posterior mean $\hat{a}$, divided by the posterior variance; the four context lengths nearly coincide once $R \gtrsim 2$. Center: squared error of the library's 16-step forecast mean relative to continuous Bayes, divided by the predictive variance. Right: ratio of error when each history is given another history's library size (same total budget) to error under matched assignment; values above one favor matching. Points average 32 libraries per history; bands are 95% bootstrap intervals over histories.

preregistered[1] primary outcome is the squared error of the library's posterior mean of a; forecast errors at horizons $h \in \{1, 4, 16\}$ are secondary.

Results. When posterior-mean error is divided by posterior uncertainty and plotted against coverage, the four context lengths nearly coincide (Figure 2, left). Moving from $R = 1/4$ to 32 reduces posterior-mean error by a factor of 1,000–5,300 and 16-step forecast error by a factor of 60–1,100, depending on the context length. At coverage between 1/2 and 4, assigning library sizes by $N_2(C)$ beats a shuffled assignment with the same total budget in all 16 comparisons (four coverage levels at each of four context lengths), and in 15 of them the paired 95% interval for the difference excludes zero. At coverage 8 and above the two assignments do equally well.

4 Trained models beyond their library's coverage

Coupled linear dynamics. We train on eight-dimensional stable linear systems

$$x_{t+1} = B\,\mathrm{diag}(\lambda)B^\top x_t + \epsilon_t, \quad \lambda_i \sim \mathrm{Unif}[-0.9, 0.9], \quad \epsilon_t \sim \mathcal{N}(0, 0.1^2 I), \tag{3}$$

where one fixed random orthogonal matrix B mixes the eight modes, so every observed coordinate depends on several of them; a system is $\theta = \lambda \in \mathbb{R}^8$. Each training sequence comes from one system in a fixed library and has 17 states. An eight-layer, width-128 causal Transformer (1.59M parameters) predicts every next state. We train on libraries of $M = 256$ and $M = 4096$ systems, the smaller being the first 256 systems of the larger, for 200,000 steps and test on 1,024 new systems drawn from the prior. Because B is known, rotating by $B^\top$ splits the posterior into eight one-dimensional integrals, which gives exact continuous-Bayes and library-Bayes forecasts for every test history. We report results after the model has seen $k \in \{3, 7, 11, 15\}$ transitions. Before training we fixed a pass criterion: risk within 20% of continuous Bayes at every k.

At $M = 4096$ the model meets the criterion at every k (Figure 3); its largest risk ratio is 1.003, so the conclusion does not depend on the 1.2 threshold. At $k = 15$ its risk ratio is $1.0007\,[0.9994, 1.0021]$,

[1] Preregistrations, code, and model states: `https://github.com/i-habib/task-library-coverage`.

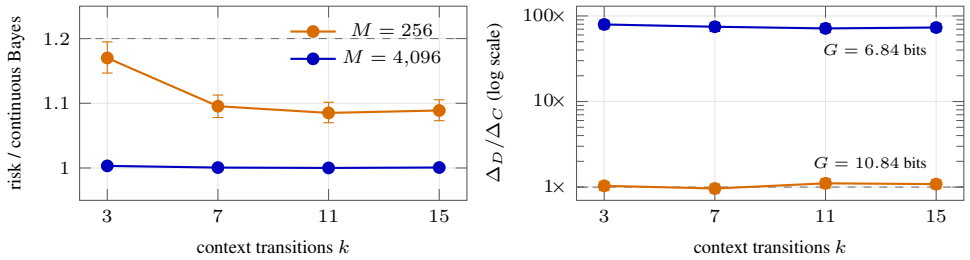


Figure 3: **Forecasting beyond the training library.** Orange: model trained on $M = 256$ systems; blue: $M = 4{,}096$. Left: next-state risk divided by continuous-Bayes risk; the dashed line is the preregistered threshold of 1.2. Right: distance ratio Δ_D/Δ_C (log scale); the dashed line at one means the model is equally close to both references. Labels give each library's shortfall G at $k = 15$ (mean over test systems). Error bars are 95% bootstrap intervals over test systems.

Model	risk / Bayes risk	Δ_D/Δ_C	shortfall G (bits)
Transformer	1.050 [1.033, 1.068]	75.0	11.42
GRU	1.301 [1.254, 1.351]	10.7	11.42

Table 1: **Transformer and GRU on in-context regression at $k = 15$, $M = 4096$.** One paired training seed; brackets are 95% bootstrap intervals over test prompts. All context lengths are in Table 2.

while the library is $G = 6.84\,[6.69, 7.00]$ bits short: lookup would need about $2^{6.84} \approx 115$ times as many systems, roughly 470,000. The predictions are 73 times closer to continuous Bayes than to library Bayes. A Bayesian lookup over the 4,096 systems the model saw is therefore incompatible with its forecasts. The $M = 256$ model, whose 256 systems are a subset of the 4,096, also forecasts well: at $k = 15$ its risk is 8.9% above continuous Bayes. But its predictions are about equally close to both references (distance ratio 1.08), and its library is 10.84 bits short. Going from 256 to 4,096 training systems changes which rule the model follows, not only how accurate it is.

In-context linear regression. We repeat the test in the setting of Raventós et al. [2023]. A prompt has 16 pairs (x_i, y_i) with $x_i \sim \mathcal{N}(0, I_8)$, $y_i = w^\top x_i + 0.5\,\epsilon_i$ with $\epsilon_i \sim \mathcal{N}(0, 1)$, and task vector $w \sim \mathcal{N}(0, I_8)$; training uses a fixed library of M task vectors. Continuous Bayes is ridge regression and library Bayes is their dMMSE. We use their eight-layer, width-128 Transformer and evaluate on 4,096 unseen tasks after $k \in \{3, 7, 11, 15\}$ observed pairs. With $M = 4096$ and $k = 15$, the library is 11.42 bits short. Across three training seeds per library size, the number of models meeting the same 20% criterion at every k rises sharply: none at $M = 1024$, two at $M = 2048$, and all three at $M = 4096$.

Transformer versus GRU. We trained a four-layer GRU (hidden size 256, 1.48M parameters) on the same 4,096-task library with the same optimizer, learning-rate schedule, and number of updates, and evaluated it on the same prompts. Both models are far closer to continuous Bayes than to library Bayes, but only the Transformer comes close to Bayes risk (Table 1). The GRU has moved away from lookup without learning the continuous estimator accurately. We did not tune the GRU's learning rate separately, so part of this gap may come from optimization rather than architecture.

5 Discussion and limitations

The test rules out lookup but does not identify what the model computes instead. Gradient-descent and ridge-like updates both fit this behavior in related settings [Akyürek et al., 2023, von Oswald et al., 2023]; telling them apart would require looking inside the network.

All systems here are simulated and linear-Gaussian, with a known prior and exact likelihood. Computing N_2 needs both, so the test cannot be applied directly to a foundation model trained on real series without a simulator whose task distribution matches the pretraining mixture. The coupled-dynamics and GRU results each use one training seed; intervals reflect variation over test systems only.

References

Sergios Agapiou, Omiros Papaspiliopoulos, Daniel Sanz-Alonso, and Andrew M. Stuart. Importance sampling: Intrinsic dimension and computational cost. *Statistical Science*, 32(3):405–431, 2017. doi: 10.1214/17-STS611.

Ekin Akyürek, Dale Schuurmans, Jacob Andreas, Tengyu Ma, and Denny Zhou. What learning algorithm is in-context learning? investigations with linear models. In *International Conference on Learning Representations*, 2023.

Abdul Fatir Ansari, Lorenzo Stella, Caner Turkmen, Xiyuan Zhang, Pedro Mercado, Huibin Shen, Oleksandr Shchur, Syama Sundar Rangapuram, Sebastian Pineda Arango, Shubham Kapoor, Jasper Zschiegner, Danielle C. Maddix, Hao Wang, Michael W. Mahoney, Kari Torkkola, Andrew Gordon Wilson, Michael Bohlke-Schneider, and Yuyang Wang. Chronos: Learning the language of time series. *Transactions on Machine Learning Research*, 2024.

Sourav Chatterjee and Persi Diaconis. The sample size required in importance sampling. *The Annals of Applied Probability*, 28(2):1099–1135, 2018. doi: 10.1214/17-AAP1326.

Abhimanyu Das, Weihao Kong, Rajat Sen, and Yichen Zhou. A decoder-only foundation model for time-series forecasting. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235, pages 10148–10167, 2024.

Shivam Garg, Dimitris Tsipras, Percy Liang, and Gregory Valiant. What can transformers learn in-context? a case study of simple function classes. In *Advances in Neural Information Processing Systems*, volume 35, 2022.

Alex Nguyen and Gautam Reddy. Differential learning kinetics govern the transition from memorization to generalization during in-context learning. In *International Conference on Learning Representations*, 2025.

Allan Raventós, Mansheej Paul, Feng Chen, and Surya Ganguli. Pretraining task diversity and the emergence of non-Bayesian in-context learning for regression. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

Johannes von Oswald, Eyvind Niklasson, Ettore Randazzo, João Sacramento, Alexander Mordvintsev, Andrey Zhmoginov, and Max Vladymyrov. Transformers learn in-context by gradient descent. In *Proceedings of the 40th International Conference on Machine Learning*, 2023.

Sang Michael Xie, Aditi Raghunathan, Percy Liang, and Tengyu Ma. An explanation of in-context learning as implicit Bayesian inference. In *International Conference on Learning Representations*, 2022.

A Supporting derivations and implementation details

A.1 Proof of the coverage identities

Write $Z_\beta(C) = \int L_C(\theta)^\beta \, dP(\theta)$, so Z_1 is the marginal likelihood of C and $N_2 = Z_2/Z_1^2$. Conditional on context C, the variables $L_C(\theta_j)$ are iid with mean Z_1 and second moment Z_2. Therefore

$$\mathrm{Var}\left[M^{-1}\sum_j L_C(\theta_j)\right] = M^{-1}(Z_2 - Z_1^2),$$

which gives the first identity in Eq. (2). Moreover,

$$\frac{\mathrm{ESS}_M}{M} = \frac{(M^{-1}\sum_j L_C(\theta_j))^2}{M^{-1}\sum_j L_C(\theta_j)^2} \xrightarrow{a.s.} \frac{Z_1^2}{Z_2} = N_2^{-1}$$

by the strong law whenever $Z_2 < \infty$. Equation (1) follows from the definitions $D_2(Q\|P) = \log \int (dQ/dP)^2 \, dP$ and $\chi^2(Q\|P) = \int (dQ/dP)^2 \, dP - 1$ with $dQ_C/dP = L_C/Z_1$.

A.2 Temporal experiment details

The Gaussian AR(1) likelihood depends on $S_{xx} = \sum_{t=0}^{n-1} x_t^2$ and $S_{xy} = \sum_{t=0}^{n-1} x_t x_{t+1}$. The continuous posterior density is proportional to $\exp(aS_{xy} - a^2 S_{xx}/2)$ on $[-0.95, 0.95]$. We used composite 96-node Gauss–Legendre quadrature with mode-centered breaks and checked every context at order 192. Given a and the final observed state x_n, the h-step conditional mean and variance are $a^h x_n$ and $\sum_{j=0}^{h-1} a^{2j}$. Posterior predictive densities integrate the full atom-specific mixture rather than substituting the posterior mean of a. Eight independent future paths per context were paired across predictors. Direct 60,000-draw mixture simulations agreed with analytic moments within 1.08 standard errors (precommitted tolerance: eight).

For the coverage intervention, we generated 256 new contexts at each $n \in \{16, 64, 256, 1024\}$. For every context and target $R \in \{1/4, 1/2, 1, 2, 4, 8, 16, 32\}$, we set $M = \lceil RN_2(C) \rceil$ and drew 32 independent prior libraries. Nested prefixes pair all coverage levels. We average library draws within each context before bootstrapping the 256 contexts (2,000 replicates). The shuffled control uses a fixed derangement of N_2 within each n, preserving library budgets while breaking their context match.

A.3 Learned coupled-dynamics experiment details

The common asset file fixes the orthogonal basis, a nested 4,096-system library, and 1,024 unseen evaluation systems before optimization. The $M = 256$ library is its literal prefix. Each system generates 17 states with $x_0 \sim \mathcal{N}(0, I)$, and training minimizes one-step error at all 16 positions. The Transformer has eight layers, width 128, two heads, and MLP width 512 (1.59M parameters). Adam uses batch size 256, linear warmup to learning rate 0.001 at step 100,000, then linear decay to the frozen 200,000 endpoint.

In modal coordinates, the likelihood factorizes across eight coefficients. Continuous posterior means and N_2 use 96-node one-dimensional Gauss–Legendre quadrature. Repeating a fixed subset at order 128 changes predictions by at most 7.64×10^{-10} and $\log_2 N_2$ by 3.14×10^{-4}. Finite Bayes weights every member of the literal library. A direct broadcast implementation and the sufficient-statistic implementation agree to 3.5×10^{-15}. All intervals use 2,000 paired-prompt bootstrap replicates. They quantify variation over unseen systems, not over training seeds. The two library sizes each use one training seed. The preregistered criteria were written in terms of $S = (\Delta_D - \Delta_C)/(\Delta_D + \Delta_C) = (r-1)/(r+1)$, where $r = \Delta_D/\Delta_C$ is the distance ratio reported in the main text, so $S > 0$ exactly when $r > 1$. The strongest preregistered outcome requires a model/Bayes risk ratio below 1.2 at all four context lengths, a positive 95% lower bound for S, and a 95% lower bound above three bits for G at $k = 15$. At $k = 15$, $S = 0.973\,[0.970, 0.976]$ for $M = 4096$.

A.4 Matched Transformer–GRU comparison

The regression comparison uses the same 4,096 task vectors, 4,096 unseen prompts, checkpoints at step 290,000, and exact continuous and discrete Bayes references for both architectures. The eight-layer Transformer has 1.59M parameters; the four-layer width-256 GRU has 1.48M. At $k = 15$, the Transformer risk ratio is 1.050 [1.033, 1.068], versus 1.301 [1.254, 1.351] for the GRU. Their S values are 0.974 [0.971, 0.976] and 0.830 [0.815, 0.842], respectively. These paired-prompt intervals describe one completed matched seed. The separate Transformer diversity sweep uses three seeds per library size and reports only fixed-threshold pass counts in the main text. Table 2 gives every context length. The GRU fails the 1.2 criterion at $k = 11$ and $k = 15$. At $k = 3$ the library covers the posterior ($G < 0$), so lookup and continuous Bayes nearly agree there.

A.5 AI assistance disclosure

An AI coding assistant monitored detached Colab jobs and handled snapshot downloads and verified restarts after runtime reclamation. Scientific choices and decision rules were fixed before outcomes; all reported measurements were computed from saved model outputs. An AI assistant also helped edit the text of this camera-ready version for clarity.

Model	k	risk / Bayes risk	Δ_D/Δ_C	G (bits)
Transformer	3	1.002 [0.997, 1.007]	14.9	-4.31
	7	1.024 [1.013, 1.036]	37.1	3.88
	11	1.059 [1.043, 1.075]	56.6	8.71
	15	1.050 [1.033, 1.068]	75.0	11.42
GRU	3	1.010 [1.002, 1.018]	6.5	-4.31
	7	1.104 [1.080, 1.129]	8.2	3.88
	11	1.218 [1.178, 1.259]	9.0	8.71
	15	1.301 [1.254, 1.351]	10.7	11.42

Table 2: Transformer and GRU on in-context regression, $M = 4096$, one paired seed, all context lengths. Brackets are 95% bootstrap intervals over the 4,096 test prompts.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and Sections 1 and 5 distinguish the exact diagnostic, simulator evidence, neural evidence, and explicit non-universality limitations.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5 states that all systems are simulated linear-Gaussian with a known prior and likelihood, that the test therefore cannot be applied directly to models trained on real series, that seed counts are small, and that the test rules out lookup without identifying the mechanism.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be

used reliably to provide closed captions for online lectures because it fails to handle technical jargon.

- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [Yes]

   Justification: Section 2 states the finite-second-moment condition implicitly through finite Z_2, and Appendix A proves both exact identities. All experiments use exact finite-context values of N_2.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Sections 3–4 and Appendix A specify distributions, grids, architecture, optimization, evaluation, uncertainty, fixed gates, and numerical tolerances. Code and model states are released (see the footnote in Section 3).

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.

- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Code, preregistrations, synthetic data, and trained model states are released at `https://github.com/i-habib/task-library-coverage`. The paper and appendix specify the distributions, model, optimization, estimators, uncertainty procedure, and decision rules needed to reproduce the main results.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Sections 3–4 give the main settings and Appendix A gives the remaining quadrature, training, and uncertainty details.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.

- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Figures 2–3 and the tables use percentile 95% bootstrap intervals with the resampling units and replicate counts stated. Seed counts are explicitly descriptive rather than confidence intervals.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [No]

   Justification: The experiments used CPU workers for the coverage intervention and NVIDIA L4/T4 GPUs for neural training. Exact aggregate GPU-hours were not retained across reclaimed cloud sessions.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
   - The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

Answer: [Yes]

Justification: The study uses only synthetic data and open research code, involves no human subjects, and follows the NeurIPS Code of Ethics.

Guidelines:

- The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
- If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
- The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: This is a foundational diagnostic study on synthetic data. Its intended benefit is more reliable evaluation; it does not release a deployable model or introduce a direct application-specific risk.

Guidelines:

- The answer [N/A] means that there is no societal impact of the work performed.
- If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
- Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
- The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
- The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
- If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The code and synthetic-array release contains no personal, scraped, hazardous, or high-risk model assets.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.

- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: [Yes]

    Justification: The Raventós et al. implementation is cited and used at a pinned commit under its Apache-2.0 license; the official NeurIPS template is used unmodified.

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: [Yes]

    Justification: Appendix A documents synthetic-data generation, evaluation, and uncertainty procedures. No consent is required for synthetic data.

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The work contains no crowdsourcing or human-subject research.

    Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: The work contains no human participants or identifiable human data.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: Appendix A discloses that an AI coding assistant monitored detached Colab jobs and handled snapshot downloads and verified restarts. Scientific choices and decision rules were fixed before outcomes, and reported measurements were computed from saved model outputs.

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.